7 Waveguides

INTRODUCTION

- A waveguide is a structure that channels waves, such as electromagnetic or sound waves, with the least amount of energy loss by limiting energy transmission to one direction.
- The waveguides are mostly hollow metal tubes. The cross-section is usually rectangular or circular in shape. Waveguides can route power precisely where it's needed, and it can withstand a large amount of power, and it functions as a high pass filter (HPF).
- Waveguide generally acts as HPF (high pass filter) where the energies higher than the specific frequencies (often referred to as the cut-off frequency) will be allowed to pass through it.
- The waveguides will reject the frequency ranging below the cut-off frequency.
- Microwave frequencies (often larger than 300 MHz or 8 GHz and above) are frequently transmitted through waveguides.
- Waveguides are commonly employed in microwave communications, radar installations, and broadcasting communications.

Key features of waveguides:

1) A waveguide is a hollow metallic tunnel that transports microwave energies from one location to another.

2) A metallic waveguide (rectangular or circular cross-section) has a single enclosed conductor filled with an insulating medium, whereas a dielectric waveguide has numerous dielectrics.

3) In the waveguide analysis, generally, the wave theory is considered.

4) Waveguides offer a higher capability for power handling.

Grey Matter Alert!

The size of a waveguide is also determined by the frequency of the transmitted wave.
Each waveguide has a cut-off the wavelength that is specified by its size and prevents waves of longer wavelengths from passing through. For example, a light-guiding optical fibre will not carry microwaves with a significantly longer wavelength.



Rack Your Brain

Name any naturally occurring structures that can also act as waveguides.

5) The operating frequency range is 3 GHz to 100 GHz in waveguides.
6) The operating modes are TE and TM modes and do not support TEM mode.
7) The larger surface area of the waveguides helps reduce copper losses.
8) Dielectric losses are less in waveguides in comparison to others.
9) Waveguides have lower signal attenuation at high frequencies.
10) The characteristic impedance (or wave impedance) is a function of frequency.
11) Waveguides act as a high pass filter.

Benefits of waveguides:

1) The power losses are generally negligible in waveguides.
2) The microwave energies while travelling through the waveguides experience lower losses than others.
3) The waveguides can easily handle immense power (in kW).
4) Waveguides are usually easy to manufacture.
5) Signal attenuation is lower in the case of waveguides.
6) Bandwidth is high as the waveguides works on high frequencies.



Rack Your Brain

A waveguide is a:
A) LPF
B) HPF
C) BPF
D) Band stop filter

Grey Matter Alert!

It is one of the lowest loss transmission line types and is highly desired for high frequency applications because most other forms of transmission structures introduce considerable losses due to energy propagation in primarily air or space within the waveguide.

S.No.	Parameters	Waveguide	Transmission Lines
1)	**Description**	A hollow metallic structure which is used to carry microwave energy from source to sink.	It is a conductor or wire which is designed to carry electrical energy below microwave range from one place to another.
2)	**Structure**	Metal waveguides are enclosed conductor filled with an insulating medium whereas dielectric waveguides have multiple dielectrics.	Transmission lines are two or more conductors separated by some insulating medium (e.g., two-wire, co-axial etc.).
3)	**Supporting theory**	Wave theory	Circuit theory
4)	**Power-handing capacity**	High	Low
5)	**Operating Frequency**	The operating frequency is 3 GHz to 100 GHz in waveguides.	The operating frequency is up to 18 GHz.
6)	**Operating mode**	TE or TM modes. (Cannot support TEM mode).	TEM mode or Quasi-TEM mode (Cannot support TE or TM modes).
7)	**Copper losses**	The larger surface area of the waveguide reduces the copper loss.	Two-wire transmission lines have smaller surface area & hence has large copper losses.
8)	**Dielectric losses**	Less dielectric losses in waveguides.	There are high dielectric losses.
9)	**Signal attenuation**	Lower signal attenuation at high frequencies.	Significant signal attenuation at high frequencies. Because of dielectric and conductor losses.
10)	**Filter functionality**	Waveguides act as a HPF.	All frequencies pass through transmission lines.

S.No.	Parameters	Waveguide	Transmission Lines
11)	**Impedance**	Wave impedance (Characteristic impedance) is a function of frequency	Characteristic impedance depends on the physical parameters of transmission lines.
12)	**Type of signal**	Electromagnetic signals are transmitted	The electrical signal is transmitted.

Table 7.1 Basic Comparisons Between Waveguides and Transmission Lines

GUIDING AND DISPERSION

Waveguide equation:

For electric field,

$$\nabla^2\vec{E}_s - \gamma^2\vec{E}_s = 0 \qquad \ldots(7.1)$$

and $\gamma^2 = j\omega\mu\left(\sigma + j\omega \in\right)$ $\ldots(7.2)$

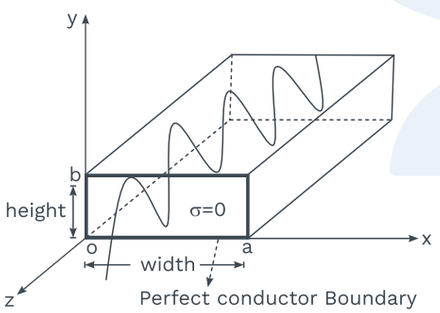


Fig. 7.1 A Hollow Rectangular Waveguide Placed Along z-axis

- Consider a hollow rectangular waveguide with dimensions a×b, placed along the z-axis direction.
  We consider:
  a = (broader dimension)
  b = (shorter dimension)

  $$\text{Aspect ratio} = \frac{\text{width}}{\text{height}}$$

Here, Aspect Ratio = $\frac{a}{b}$

A standard rectangular waveguide has $a \geq 2b$.

Grey Matter Alert!

The boundary conditions are:

- Electromagnetic waves do not pass through conductors but are rather reflected.
- Any electric field that touches a conductor must be perpendicular to it.
- Any magnetic field close to a conductor must be parallel to it.

- If $\sigma = 0$ (for a lossless dielectric medium) where wave travels, then:

$$\gamma^2 = j^2\omega^2\mu \in$$

$$\gamma^2 = -\omega^2\mu \in = -K^2 \quad \{\because K^2 = \omega^2\mu \in\}$$

$$\nabla^2\vec{E}_s + K^2\vec{E}_s = 0 \qquad ...(7.3)$$

The above equation 7.3) is often called as Helmholtz equation.

Wave number, $K = \omega\sqrt{\mu \in}$ (for free space) ...(7.4)

- For Magnetic Field:

$$\nabla^2\vec{H}_s + K^2\vec{H}_s = 0 \qquad ...(7.5)$$

- The components of vectors $\vec{E}_s$ and $\vec{H}_s$ will be as follows:

$\vec{E}_s(E_{xs}, E_{ys}, E_{zs})$ and $\vec{H}_s(H_{xs}, H_{ys}, H_{zs})$

- These six equations will help in defining the electric field and magnetic field properly.

Now, consider the wave is propagating in the z-direction.

For z-component only:

The electric field will be:

$$\frac{\partial^2 E_{zs}}{\partial x^2} + \frac{\partial^2 E_{zs}}{\partial y^2} + \frac{\partial^2 E_{zs}}{\partial z^2} + K^2 E_{zs} = 0$$

(Partial differential equation)

Using separation of variables method:

$$E_{zs} = X(x)\, Y(y)\, Z(z) \qquad ...(7.6)$$

$$\frac{\partial^2 XYZ}{\partial x^2} + \frac{\partial^2 XYZ}{\partial y^2} + \frac{\partial^2 XYZ}{\partial z^2} + K^2 XYZ = 0$$

Dividing both sides by XYZ, we get:

$$\frac{YZ}{XYZ}\frac{\partial^2 X}{\partial x^2} + \frac{XZ}{XYZ}\frac{\partial^2 Y}{\partial y^2} + \frac{XY}{XYZ}\frac{\partial^2 z}{\partial z^2} + K^2 = 0$$

$$\frac{1}{X}\frac{\partial^2 X}{\partial x^2} + \frac{1}{Y}\frac{\partial^2 Y}{\partial y^2} + \frac{1}{Z}\frac{\partial^2 Z}{\partial z^2} + K^2 = 0 \qquad ...(7.7)$$

Rack Your Brain

List some practical uses of rectangular waveguides?

Previous Years' Question

For a hollow waveguide, the axial current must necessarily be:

a) Conduction current only.

b) Time-varying conduction current and displacement current.

c) A combination of conduction and displacement current.

d) Displacement current only.

$$-K_x^2 - K_y^2 + \gamma^2 = -K^2 \quad \ldots(7.8)$$

where

$-K_x^2, -K_y^2, \gamma^2$ are separation constants.

$$\frac{1}{X}\frac{\partial^2 X}{\partial x^2} = -K_x^2 \Rightarrow \frac{1}{X}\frac{\partial^2 X}{\partial x^2} + K_x^2 = 0$$

$$\frac{\partial^2 X}{\partial x^2} + K_x^2 X = 0 \quad \ldots(7.9)$$

The equation 7.9) is a form of partial differentiation.

$$D^2 + K_x^2 = 0$$

$$D = \pm jK_x$$

$$\therefore \quad X(x) = C_1 \cos K_x x + C_2 \sin K_x x \quad \ldots(7.10)$$

- Similarly,

$$\frac{1}{Y}\frac{\partial^2 Y}{\partial y^2} = -K_y^2$$

$$\frac{1}{Y}\frac{\partial^2 Y}{\partial y^2} + K_y^2 = 0$$

$$\frac{\partial^2 Y}{\partial y^2} + y K_y^2 = 0$$

$$\therefore \quad Y(y) = (C_3 \cos K_y y + C_4 \sin K_y y) \quad \ldots(7.11)$$

Also, $\frac{1}{Z}\frac{\partial^2 Z}{\partial z^2} = \gamma^2$

$$\frac{1}{Z}\frac{\partial^2 Z}{\partial z^2} - \gamma^2 = 0$$

$$\frac{\partial^2 Z}{\partial z^2} - \gamma^2 Z = 0$$

- The above equation is in the form of:
  $D^2 - \gamma^2 = 0$ (Complementary function)
  $D = \pm\gamma$

$$\therefore \quad Z(z) = (C_5 e^{-\gamma z} + C_6 e^{+\gamma z}) \quad \ldots(7.12)$$

Substituting equations 7.10), 7.11) and 7.12), in equation 7.6).

Previous Years' Question

Consider a standard rectangular and a circular waveguide having the same dominant mode cut-off frequency. The ratio of the areas of a circular waveguide to a rectangular waveguide.

a) π **b)** $\frac{1.84}{\pi}$

c) $\frac{0.69}{\pi}$ **d)** $\frac{1}{\pi}$

Grey Matter Alert!

In the medical test ultrasonography, the waveguide is put in contact with the specimen or the specimen may be put inside the waveguide (as in a dielectric constant measurement) where the waveguide ensures that the power of the testing wave is conserved so that smaller objects are tested, and the accuracy is enhanced.

$E_{zs} = (C_1 \cos K_x x + C_2 \sin K_y y)(C_3 \cos K_y y + C_4 \sin K_y y)(C_5 e^{-\gamma z} + C_6 e^{\gamma z})$

- If the wave is travelling in positive z-direction:

$E_{zs} = (C_1 \cos K_x x + C_2 \sin K_x x)(C_3 \cos K_y y + C_4 \sin K_y y)(C_5 e^{-\gamma z})$

$E_{zs} = (A \cos K_x x + B \sin K_x x)(C \cos K_y y + D \sin K_y y) e^{-\gamma z}$ ...(7.13)

- Similarly,

$H_{zs} = (P \cos K_x x + Q \sin K_x x)(R \cos K_y y + S \sin K_y y) e^{-\gamma z}$

Now calculating E_{xs}, E_{yz}, H_{xs}, H_{ys}

We will use the Maxwell's equations.

- The Maxwell equation is given by:

$$\nabla \times \vec{E}_s = -j\omega\mu\vec{H}_s \quad \text{...(7.14)}$$

here, 's' denotes that the time factor is suppressed.

From equation 7.14)

$$\nabla \times \vec{E}_s = -j\omega\mu(H_{xs}\hat{a}_x + H_{ys}\hat{a}_y + H_{zs}\hat{a}_z) = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial_x} & \frac{\partial}{\partial_y} & \frac{\partial}{\partial_z} \\ E_{xs} & E_{ys} & E_{zs} \end{vmatrix}$$

Comparing x, y, z components on both sides in the above equation

$$-j\omega\mu H_{xs} = \left[\frac{\partial E_{zs}}{\partial y} - \frac{\partial E_{ys}}{\partial z}\right] \quad \text{...(7.15)}$$

$$-j\omega\mu H_{ys} = -\left[\frac{\partial E_{zs}}{\partial x} - \frac{\partial E_{xs}}{\partial z}\right] \quad \text{...(7.16)}$$

$$-j\omega\mu H_{zs} = \left[\frac{\partial E_{ys}}{\partial x} - \frac{\partial E_{xs}}{\partial y}\right] \quad \text{...(7.17)}$$

- The Maxwell equation for $\vec{H}$

$$\nabla \times \vec{H}_s = (\sigma + j\omega\in)\vec{E}$$

As $\sigma = 0$ (loss less dielectric)

$$\therefore \nabla \times \vec{H}_s = j\omega\in\left[E_{xs}\hat{a}_x + E_{ys}\hat{a}_y + E_{zs}\hat{a}_z\right]$$

$$= \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_{xs} & H_{ys} & H_{zs} \end{vmatrix}$$

Rack Your Brain

What are acoustic waveguides?

- Comparing x, y, z components on both sides.

$$j\omega \in E_{xs} = \left[\frac{\partial H_{zs}}{\partial y} - \frac{\partial H_{ys}}{\partial z}\right] \quad ...(7.18)$$

$$j\omega \in E_{ys} = -\left[\frac{\partial H_{zs}}{\partial x} - \frac{\partial H_{xs}}{\partial z}\right] \quad ...(7.19)$$

$$j\omega \in E_{zs} = \left[\frac{\partial H_{ys}}{\partial x} - \frac{\partial H_{xs}}{\partial z}\right] \quad ...(7.20)$$

- Now, to get the values, we substitute from equation 7.18) and 7.16)

$$j\omega \in E_{xs} = \frac{\partial H_{zs}}{\partial y} - \frac{1}{j\omega\mu}\frac{\partial}{\partial z}\left(\frac{\partial E_{zs}}{\partial x} - \frac{\partial E_{xs}}{\partial z}\right)$$

$$j\omega \in E_{xs} = \frac{\partial H_{zs}}{\partial y} - \frac{1}{j\omega\mu}\frac{\partial^2 E_{zs}}{\partial x \partial z} + \frac{1}{j\omega\mu}\frac{\partial^2 E_{xs}}{\partial z^2} \quad ...(7.21)$$

- If the wave is propagating in z-direction, then it will have some peak value and gets attenuated:

$E_{zs} = E_{z0}\, e^{-\gamma z}$, $E_{xs} = E_{x0} e^{-\gamma z}$

Here, E_{z0} is the peak value at z = 0 and its component will vary in the propagation direction (x-direction also).

$$E_{xs} = E_{x0} e^{-\gamma z}$$

$$\therefore \quad \frac{\partial E_{xs}}{\partial z} = -\gamma E_{x0} e^{-\gamma z}$$

$$\frac{\partial^2 E_{xs}}{\partial z^2} = \gamma^2 E_{x0} e^{-\gamma z} = \gamma^2 E_{xs}$$

Similarly, $E_{zs} = E_{z0} e^{-\gamma z}$

$$\frac{\partial E_{zs}}{\partial z} = -\gamma E_{z0} e^{-\gamma z} = -\gamma E_{zs}$$

- Substituting the above values in equation 7.21), we get:

$$j\omega \in E_{xs} = \frac{\partial H_{zs}}{\partial y} - \frac{1}{j\omega\mu}\frac{\partial E_{zs}}{\partial x}(-\gamma) + \frac{1}{j\omega\mu}\gamma^2 E_{xs}$$

Rack Your Brain

_____ acts as the equivalent of cables for super high frequency (SHF) systems.

- Rearranging the equation:

$$j\omega \in E_{xs} - \frac{\gamma^2}{j\omega\mu} E_{xs} = \frac{\gamma}{j\omega\mu}\frac{\partial E_{zs}}{\partial x} + \frac{\partial H_{zs}}{\partial y}$$

$$\frac{-\omega^2\mu \in E_{xs} - \gamma^2 E_{xs}}{j\omega\mu} = \frac{\gamma}{j\omega\mu}\frac{\partial E_{zs}}{\partial x} + \frac{\partial H_{zs}}{\partial y}$$

$$(\omega^2\mu \in +\gamma^2)E_{xs} = -\gamma\left(\frac{\partial E_{zs}}{\partial x}\right) - j\omega\mu\left(\frac{\partial H_{zs}}{\partial y}\right)$$

$$E_{xs} = \frac{-\gamma}{h^2}\left(\frac{\partial E_{zs}}{\partial x}\right) - \frac{j\omega\mu}{h^2}\left(\frac{\partial H_{zs}}{\partial y}\right) \quad ...(7.22)$$

where $h^2 = \omega^2\mu \in + \gamma^2 = K_x^2 + K_y^2$

h is a transverse propagation constant.

- Similarly, we get:

$$E_{ys} = \frac{-\gamma}{h^2}\left(\frac{\partial E_{zs}}{\partial y}\right) + \frac{j\omega\mu}{h^2}\left(\frac{\partial H_{zs}}{\partial x}\right) \quad ...(7.23)$$

$$H_{xs} = \frac{-\gamma}{h^2}\left(\frac{\partial H_{zs}}{\partial x}\right) - \frac{j\omega \in}{h^2}\left(\frac{\partial E_{zs}}{\partial y}\right) \quad ...(7.24)$$

$$H_{ys} = \frac{-\gamma}{h^2}\left(\frac{\partial H_{zs}}{\partial y}\right) - \frac{j\omega \in}{h^2}\left(\frac{\partial E_{zs}}{\partial x}\right) \quad ...(7.25)$$

- Since we are getting different equations for different types of fields, thus representing them we use modes.
  Any configuration of electric field and magnetic field that exists inside the waveguides must be a solution to Maxwell's equation.
  The above different equations relating to electric and magnetic fields indicate different field configurations or field patterns.
  Hence, the above equations must be a solution to Maxwell's equation.
- Besides this, the field must satisfy the boundary conditions.
  $E_{t1} = E_{t2}$
  But for the waveguide, medium-2 is a metal, a perfect conductor, and $E_{t2} = 0$ in medium-2.

$\therefore$ $E_{t1} = E_{t2} = 0$

Previous Years' Question

A rectangular-metal waveguide filled with a dielectric material of relative permittivity $\in_r = 4$ has an inner dimension of 3cm × 1.2 cm. The cut-off frequency for the dominant mode is:

a) 2.5 GHz **b)** 12.5 GHz
c) 10 GHz **d)** 5 GHz

Grey Matter Alert!

Generally, the waveguides can be designed to carry waves over a wide portion of the electromagnetic spectrum but are primarily used in the microwave and optical frequency ranges.

Different field configurations satisfy these requirements.

- Accordingly, each such configuration is called mode.

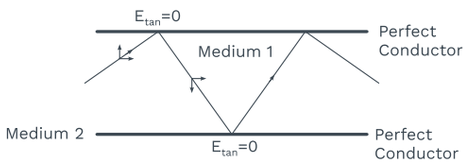


Fig. 7.2 Guiding through a Waveguide

TYPES OF MODES

1) Transverse electromagnetic mode (TEM mode):

- In this mode, the electric field, the magnetic field and the direction of propagation are perpendicular to each other.
- In the direction of wave propagation, the electric field and magnetic field will not exist.
  i.e., E_{zs}= 0 and H_{zs}= 0.
  If we substitute E_{zs}=0 and H_{zs}= 0 in equations 7.22), 7.23), 7.24) and 7.25), we get:
  $E_{xs} = E_{ys} = H_{xs} = H_{ys} = 0$ (All fields will vanish)
  That's why TEM does not exist.

2) Transverse electric mode (TE_{mn} mode):

- In this mode, the electric field will exist in the transverse (perpendicular) directions to the direction of propagation of wave only.
  Here, the considered direction of propagation of the wave is z-direction.
  Therefore, the electric field will exist in x or y-directions and will be absent in the z-direction.
  Or, $E_{zs} = 0$, and $H_{zs} \neq 0$.
- In other words, the electric field component will not be present in the direction of wave propagation, but the magnetic field will exist in the wave propagation direction.

Previous Years' Question

Assuming perfect conductors of a transmission line, pure TEM propagation is NOT possible in which of the following?

a) Co-axial cable
b) Air-filled cylindrical waveguide
c) Parallel twin-wire line in the air
d) Semi-infinite parallel plate waveguide

Grey Matter Alert!

A waveguide carries energy to and from the antenna in radar applications, where the impedance must be matched for efficient power transmission.

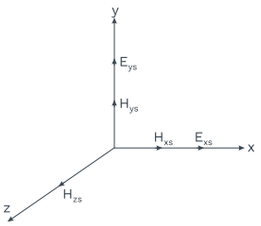


Fig. 7.3 Field Components

- TE mode is also sometimes referred to as H-mode because here in this mode the magnetic field's component is only present.

3) Transverse magnetic mode (TM_{mn} mode):

- In this mode, the magnetic field will exist in the transverse (or perpendicular) direction to the direction of propagation.
- The magnetic field will exist in x or y-direction and will be absent in z-direction i.e., $H_{zs} = 0$ but $E_{zs} \neq 0$.
- In another way, the magnetic field component will not be present in the direction of the wave propagation, but the electric field will be there in the direction of wave propagation.

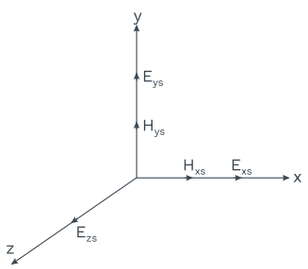


Fig. 7.4 Field Components

- This mode is sometimes referred to as E-mode because the electric field component is present in the z-direction.

Note:
These arrows in figure 7.3) and figure 7.4) are denoting only the presence of that component

Rack Your Brain

Is it true that hybrid modes have both electric and magnetic fields components in the direction of the wave propagation?

Grey Matter Alert!

By solving the Helmholtz equation and applying boundary conditions based on the geo-metrical shape and materials, the propagation modes are evaluated.

Grey Matter Alert!

By defining the region's boundaries for infinitely long uniform waveguides, the typical approach enables us to assume a propagating form for the wave, i.e., every field component has a known dependence on the propagation direction (i.e., z-direction).

in a particular direction. It is not indicating the presence of components at the same time in a specific direction.

Transverse magnetic mode:

- The condition for transverse-magnetic mode is

  $H_{zs} = 0$ and $E_{zs} \neq 0$

  And we know that,

$$E_{zs} = (A\cos K_x x + B\sin K_x x)(C\cos K_y y + D\sin K_y y)e^{-\gamma z} \quad ...(7.26)$$

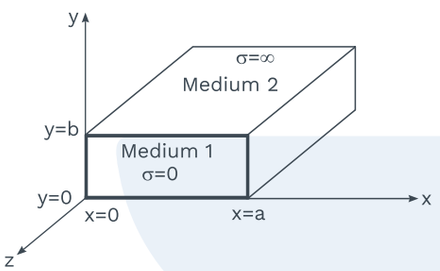


Fig. 7.5 Boundary Conditions

- Applying Boundary conditions:
  At x = 0 (y-z plane)

  $E_{tan1} = E_{tan2}$

Here, medium-1 is inside the waveguide and medium-2 is the waveguide material.

Medium-1 is lossless medium ($\sigma = 0$) whereas medium-2 is a perfect conductor ($\sigma = \infty$).

- If x = constant, then-

i) x will be normal component.

ii) y and z will be tangential components.

- For medium-1 and medium-2 tangential components are:

  $E_{y1} = E_{y2}$ and $E_{z1} = E_{z2}$

Since, medium-2 is a perfect conductor,

$E_{tan2}(E_{y2}$ and $E_{z2})$ and $E_{n2}(E_{x2})$ will be zero.

Therefore,

$E_{y2} = 0 = E_{y1}$ and $E_{z2} = 0 = E_{z1}$

As E_{z1} inside the waveguide is zero.

Previous Years' Question

The first root of $J_1(x) = 0$ and $J_1(x) = 0$ are 3.83 and 1.85 respectively. Which of the following gives the cut-off wavelength of TM_{11} mode in a circular waveguide of diameter D.

a) $\lambda_c \approx 0.64D$

b) $\lambda_c \approx 0.82D$

c) $\lambda_c \approx D$

d) $\lambda_c \approx 1.2D$

- At $x = 0$, $E_{zs} = 0$

Now, substituting values in equation 7.13)

$$0 = (A \cos K_x \times 0 + B \sin K_x \times 0)(C \cos K_y y + D \sin K_y y)e^{-\gamma z}$$

$$0 = A(C \cos K_y y + D \sin K_y y)e^{-\gamma z}$$

To satisfy the equation, A must be zero i.e. A = 0.

Then, $E_{zs} = B \sin K_x x(C \cos K_y y + D \sin K_y y)e^{-\gamma z}$

- At $y = 0$ (x-z plane), here normal component will be y and tangential component will be x and z i.e., $E_{x2} = 0$ and $E_{z2} = 0$

Substituting values in equation 7.13)

$$0 = B \sin K_x x\,[C \cos K_y \times 0 + D \sin K_y \times 0]e^{-\gamma z}$$

To satisfy, C = 0

$$\therefore \quad E_{zs} = B \sin K_x x \;\; D \sin K_y y$$

- At $x = a$, $E_{zs} = 0$

For any constant value of x, the tangential component will be zero.

Here, at x = a, the normal component will be x and the tangential component will be y and z.

$$0 = B \sin(K_x a)\; D \sin(K_y y)e^{-\gamma z}$$

- For $K_x a = m\pi$, the above equation will be satisfied.

$$\therefore \quad K_x = \frac{m\pi}{a} \qquad \ldots(7.27)$$

where m = number of half-cycle variations in x-direction.

- At $y = b$, $E_{zs} = 0$

$$0 = B \sin(K_x x)\; D \sin(K_y y)e^{-\gamma z}$$

Put, $K_x = \frac{m\pi}{a}$

$$0 = B \sin\left(\frac{m\pi}{a} x\right) D \sin(K_y b)e^{-\gamma z} \qquad \ldots(7.28)$$

here, $K_y b = n\pi$

$$\therefore \quad K_y = \frac{n\pi}{b} \qquad \ldots(7.29)$$

where n = number of half-cycle variations in y-directions.

Previous Years' Question

The longitudinal component of the magnetic field inside an air-filled rectangular waveguide made of a perfect electric conductor is given by the following expression: $H_z(x, y, z, t) = 0.1 \cos(25\pi x) \cos(30.3\pi y) \cos(12\pi \times 10^9 t - \beta z)$ (A/m).

The cross-sectional dimensions of the waveguide are given as a = 0.08 m and b = 0.033m.

The mode of propagation inside the waveguide is:

a) TM_{12} **b)** TE_{21}

c) TM_{21} **d)** TE_{12}

Substitute equations 7.27) and 7.29) in 7.28).

$$E_{zs} = E_0 \sin\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi}{b}y\right)e^{-\gamma z} \quad ...(7.30)$$

where E_0 = Peak Amplitude of the electric field.
The combined values of B and D are constant.

Here, m and n cannot be zero.

- If m = 1 and n = 1 i.e., TM_{11} mode is the minimum mode that can exist in TM_{mn} mode.

Transverse electric mode (TE_{mn} mode):

- The condition for transverse-electric mode is:

$E_{zs} = 0$ and $H_{zs} \neq 0$

And we know that,

$$H_{zs} = (P\cos K_x x + Q\sin K_x x)(R\cos K_y y + S\sin K_y y)e^{-\gamma z} \quad ...(7.31)$$

- Applying Boundary condition

At x = 0, the $E_{zs} = 0$ and $E_{ys} = 0$

From equation 7.23), we get:

$$\therefore \quad \frac{\partial H_{zs}}{\partial x} = 0$$

- At x = a, $E_{zs} = 0$ and $E_{ys} = 0$

i.e. $\frac{\partial H_{zs}}{\partial x} = 0$

- At y = 0 and y = b (x-z plane)

$E_{xs} = 0$ and $E_{zs} = 0$

From equation 7.22), we get:

$$\therefore \quad \frac{\partial H_{zs}}{\partial y} = 0$$

As $\frac{\partial H_{zs}}{\partial x} = 0$

From equation 7.31)

$$\frac{\partial H_{zs}}{\partial x} = \left[-PK_x x\sin K_x + QK_x\cos K_x x\right]\left[R\cos K_y y + S\sin K_y y\right]e^{-\gamma z}$$

$$0 = Q\,K_x[R\cos K_y y + S\sin K_y y]e^{-\gamma z} \quad ...(7.32)$$

Previous Years' Question

In a rectangular waveguide:

a) TE and TEM modes can exist, but TM mode cannot exist.
b) TE and TM modes can exist, but TEM mode cannot exist
c) TM and TEM modes can exist, but TE mode cannot exist
d) TE, TM, and TEM all modes can exist.

here, Q = 0 to satisfy above equation.
The general equation is:

$$H_z = P \cos K_x x\,[R \cos K_y y + S \sin K_y y]e^{-\gamma z} \quad \ldots(7.33)$$

- At y = 0

$$\frac{\partial H_{zs}}{\partial y} = 0$$

$$\frac{\partial H_{zs}}{\partial y} = P \cos K_x x\left[-R K_x \sin K_y y + S K_y \cos K_y y\right]e^{-\gamma z}$$

$$0 = P \cos K_x x\, S\, k_y$$

Here, S = 0 must be there to satisfy the above equation.

$$H_{zs} = P \cos(K_x x)\, R \cos(K_y y)e^{-\gamma z}$$

$$\frac{\partial H_{zs}}{\partial x} = -P \sin(K_x x) K_x R \cos(K_y y)\, e^{-\gamma z}$$

- At x = a, $\frac{\partial H_{zs}}{\partial x} = 0$

$$0 = -P \sin(K_x a)\, R \cos(K_y y)\, e^{-\gamma z}$$

As $P \neq 0$ and $K_x a = m\pi$

$$\therefore \quad K_x = \frac{m\pi}{a}$$

$$H_{zs} = P \cos\left(\frac{m\pi}{a} x\right) R \cos(K_y y)\, e^{-\gamma z}$$

- At y = b, $\frac{\partial H_{zs}}{\partial y} = 0$

$$\frac{\partial H_{zs}}{\partial y} = P \cos\left(\frac{m\pi}{a} x\right) R \sin(K_y y) K_y e^{-\gamma z}$$

$$0 = -P \cos\left(\frac{m\pi}{a} x\right) R \sin(K_y b) K_y e^{-\gamma z}$$

and $K_y b = n\pi$,

$$K_y = \frac{n\pi}{b}$$

Hence, $H_{zs} = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi}{b} y\right) e^{-\gamma z}$

Previous Years' Question

In a rectangular cavity resonator having width 'a', height 'b', and length 'd' for d > a > b, then the dominant mode is:

a) TE_{101} **b)** TM_{101}
c) TE_{110} **d)** TM_{110}

here, H_0 = Peak amplitude of the magnetic field
m & n ≠ 0 simultaneously for this mode to exist.

Example 1:
What will happen if m and n are zero?

Sol:
If both m and n are zero then, $H_{zs} = H_0$
i.e., peak amplitude or it will have no frequency and we know that waveguides act as HPF and it passes the frequencies higher than cut-off frequency.
TE_{10} and TE_{01} will be the minimum mode that can exist in TE_{mn} mode.

DISPERSION RELATION IN THE WAVEGUIDE (WAVEGUIDE PARAMETER)

- The transverse propagation constant is given by:

$$h^2 = \gamma^2 + \omega^2\mu\epsilon = K_x^2 + K_y^2$$

$$\gamma^2 = \sqrt{K_x^2 + K_y^2 - \omega^2\mu\epsilon} \quad \ldots(7.34)$$

- We know that,

$$\gamma = \alpha + j\beta$$

and from equation 7.34)

$$\gamma^2 = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \omega^2\mu\epsilon}$$

- At cut-off frequency, $\omega = \omega_c$, $\gamma = 0$

$$0 = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \omega_c^2\mu\epsilon}$$

$$\omega_c^2\mu\epsilon = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \quad \ldots(7.35)$$

$$(2\pi f_c)^2\mu\epsilon = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

Previous Years' Question

A hollow cubic cavity resonator has a dominant resonant frequency of 10 GHz. The length of each side is:

a) $\sqrt{3}$ cm **b)** $\sqrt{2}$ cm

c) $\sqrt{\frac{3}{2}}$ cm **d)** $\frac{3}{\sqrt{2}}$ cm

$$f_c^2 = \frac{1}{4\mu\in}\left[\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2\right]$$

$$f_c = \frac{1}{2\sqrt{\mu\in}}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

or $$f_c = \frac{v_p}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \quad \left\{\because\ v_p = \frac{1}{\sqrt{\mu\in}}\right\} \quad ...(7.36)$$

where v_p = phase velocity

- For air-filled waveguide, the phase velocity

  $v_p = c = 3 \times 10^8$ m/sec

- **Cut-off frequency**:

The minimum frequency above which wave propagates and below which wave attenuates.

$$f_c = \frac{c}{2}\sqrt{\left(\frac{m}{2}\right)^2 + \left(\frac{n}{b}\right)^2} \quad ...(7.37)$$

- **Cut-off wavelength:**

Maximum wavelength above which wave will attenuate and below which wave propagates. Therefore, the cut-off wavelength will be-

$$\lambda_c = \frac{v_p}{f_c} = \frac{v_p}{\frac{v_p}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}}$$

$$\lambda_c = \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}}$$

Dominant mode:

- The mode that gives the minimum cut-off frequency and maximum cut-off wavelength.
- Out of TE_{mn} modes, only TE_{00} mode does not exist but TE_{10} and TE_{01} modes exist whereas only TM_{11} mode exists out of TM_{mn} modes.

Previous Years' Question

For a rectangular waveguide of interior dimensions, a×b(a>b). The cut-off frequency for the TE_{11} mode is the arithmetic mean of the cut-off frequency for TE_{10} mode and TE_{20} mode.

If $a = \sqrt{5}$ cm, then the value of b(in cm) is _____

Previous Years' Question

The dominant mode in a waveguide is characterized by:

a) Zero attenuation.
b) Shortest frequency.
c) Longest cut-off frequency.
d) Infinite transmission.

Thus, out of all modes, the mode which satisfies the condition for dominant mode can be analyzed.

Only TE_{10} mode satisfies the necessary condition.

- Cut-off frequency for TE_{10} mode will be as follows:

$$(f_c)_{TE_{10}} = \frac{v_p}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$= \frac{v_p}{2}\sqrt{\left(\frac{1}{a}\right)^2 + 0}$$

$$\therefore \quad (f_c)_{TE_{10}} = \frac{v_p}{2a} \qquad ...(7.38)$$

- Cut-off frequency for TE_{01} mode:
  $m = 0$ & $n = 1$

$$(f_c)_{TE_{01}} = \frac{v_p}{2}\sqrt{0 + \left(\frac{1}{b}\right)^2}$$

$$(f_c)_{TE_{01}} = \frac{v_p}{2b} \qquad ...(7.39)$$

- For a standard rectangular waveguide, $a \geq 2b$

Therefore, if $a \geq 2b$, then out of TE_{01} and TE_{10} mode, the dominated mode will be TE_{10} mode as it will give the minimum cut-off frequency.

Generally, TE_{10} mode is the dominant mode.

Note:

Always select the broader dimension as a for solving the question, because of this assumption $a \geq 2b$ condition will be satisfied & by default TE_{10} will become the dominant mode.

- The cut-off wavelength for TE_{10} mode is given as:

$$\lambda_c = \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}}$$

Previous Years' Question

Evanescent mode attenuation in a waveguide depends upon the:

A) Standing wave in the guide.
B) Operated frequency.
C) Conductivity of the guide walls.
D) Conductivity of the dielectric filling the waveguide.

$$(\lambda_c)_{TE_{10}} = \frac{2}{\sqrt{\left(\frac{1}{a}\right)^2 + 0}} = 2a \qquad ...(7.40)$$

$$\gamma = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \omega^2\mu\in}$$

- And we know that, from the equation 7.35)

$$\omega_c^2\mu\in = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

$$\therefore\ \gamma = \sqrt{\omega_c^2\mu\in - \omega^2\mu\in} = \alpha + j\beta \qquad ...(7.41)$$

where α = signal attenuation and β = phase constant

	γ	α	β	**Wave propagation**	
$\omega = \omega_c$	0	0	0	Non-propagating mode	} Evanescent Mode
$\omega < \omega_c$	Real quantity	$\alpha \neq 0$	$\beta = 0$	Non-propagating mode	
$\omega > \omega_c$	Imaginary quantity	$\alpha = 0$	$\beta \neq 0$	Propagating mode	

Table 7.2 Wave Propagation

Guided phase constant (β_g):

As we know that:

$$\beta = \frac{2\pi}{\lambda}$$

- For phase constant $= \dfrac{\text{Phase shift}}{\text{Unit distance}}$
- If a wave travels in a waveguide (i.e., guided medium), then β will be referred to as guided phase constant.

whereas the phase constant outside the waveguide is the constant of a plane EM wave in free space & denoted by phase constant (β).

Previous Years' Question

The degenerate modes in a waveguide are characterized by:

a) Different cut-off frequencies and different field distribution.
b) Same cut-off frequencies but different field distributions
c) Different cut-off frequencies but same field distributions.
d) Same cut-off frequencies and same field distributions.

- For the guided phase constant:

$\gamma = \alpha_g + j\beta_g$

Here, wave is inside the waveguide i.e., guided medium

Also, $\gamma = j\beta_g = \sqrt{\omega_c^2 \mu \in -\omega^2 \mu \in}$

Since propagating mode (from Table-7.2)

$\alpha_g = 0$ and $\beta_g \neq 0$

Thus, $j\beta_g = \sqrt{(-1)\left[\omega^2 \mu \in -\omega_c^2 \mu \in\right]}$

- On comparing both sides:

$$\beta_g = \sqrt{\omega^2 \mu \in \left(1 - \frac{\omega_c^2 \mu \in}{\omega^2 \mu \in}\right)}$$

$$\beta_g = \omega\sqrt{\mu \in}\sqrt{1 - \left(\frac{\omega_c^2}{\omega}\right)}$$

$$\beta_g = \omega\sqrt{\mu \in}\sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

or $$\beta_g = \beta\sqrt{1 - \left(\frac{f_c}{f}\right)^2} \quad ...(7.42)$$

where β_g = phase constant for guided medium and β = phase constant in the free space of plane EM wave.

Guided wavelength (λ_g):

- As we know that-

$$\lambda_g = \frac{2\pi}{\beta_g}$$

$$\lambda_g = \frac{2\pi}{\beta\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$$

$$\lambda_g = \frac{\lambda}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} \quad ...(7.43)$$

Previous Years' Question

Consider an air-filled rectangular waveguide with dimensions a = 2.286 cm and b = 1.016 cm. At 10 GHz operating frequency, the value of the propagation constants (per meter) of the corresponding propagation mode is _____

a) TE_{10} **b)** TE_{01}
c) TM_{10} **d)** TM_{01}

where λ_g = guided wavelength and λ = wavelength of plane EM wave.
From equation 7.43)

$$\lambda_g = \frac{\lambda}{\sqrt{1-\left(\frac{\lambda}{\lambda_c}\right)^2}}$$

$$\lambda_g^2 = \frac{\lambda^2}{1-\left(\frac{\lambda}{\lambda_c}\right)^2}$$

$$1-\frac{\lambda^2}{\lambda_c^2} = \frac{\lambda^2}{\lambda_g^2}$$

$$\frac{1}{\lambda^2}-\frac{1}{\lambda_c^2} = \frac{1}{\lambda_g^2}$$

or $$\frac{1}{\lambda^2} = \frac{1}{\lambda_c^2}+\frac{1}{\lambda_g^2} \quad ...(7.44)$$

Guided phase velocity (v_{PZ}):

- For the plane waves,

$$v_p = \frac{\omega}{\beta}$$

- For inside the waveguide, the guided phase velocity (v_{PZ}) is

$$v_{pz} = \frac{\omega}{\beta\sqrt{1-\left(\frac{f_c}{f}\right)^2}}$$

$$v_{pz} = \frac{v_p}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}} \quad ...(7.45)$$

Guided group velocity (v_g):

- As we know,

$$v_g = \frac{\partial\omega}{\partial\beta_g}$$

Previous Years' Question

Consider an air-filled rectangular waveguide with dimensions a = 2.286 cm and b = 1.016 cm. The increasing order of the cut-off frequencies for different mode is:

a) $TE_{10} < TE_{20} < TE_{01} < TE_{11}$
b) $TE_{01} < TE_{10} < TE_{11} < TE_{20}$
c) $TE_{20} < TE_{11} < TE_{10} < TE_{01}$
d) $TE_{10} < TE_{11} < TE_{20} < TE_{01}$

$$\beta_g = \beta\sqrt{1-\left(\frac{f_c}{f}\right)^2} = \sqrt{\omega^2\mu\in - \omega_c^2\mu\in}$$

$$\frac{\partial\beta_g}{\partial\omega} = \frac{1}{2\sqrt{\omega^2\mu\in - \omega_c^2\mu\in}} \times 2\omega(\mu\in)$$

$$\frac{\partial\beta_g}{\partial\omega} = \frac{1}{\omega\sqrt{\mu\in}\sqrt{1-\left(\frac{\omega_c}{\omega}\right)^2}} \times \omega(\mu\in)$$

$$\frac{\partial\beta_g}{\partial\omega} = \frac{\sqrt{\mu\in}}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}}$$

$$v_g = \frac{\partial\omega}{\partial\beta_z} = v_p\sqrt{1-\left(\frac{f_c}{f}\right)^2}$$

Since, $v_p = \frac{1}{\sqrt{\mu\in}}$

- The phase velocity for guided medium:

$$v_{pz} = \frac{v_p}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}}$$

and $v_p^2 = v_{pz}v_g$

$$v_p^2 = \sqrt{v_{pz}v_g} \qquad ...(7.46)$$

here, v_p = phase velocity of plane EM wave,

v_{pz} = Guided phase velocity

and v_g = Guided group velocity

Example 2:
A K-band waveguide with the dimensions of 1.073 cm × 0.75 cm is filled by a dielectric medium with $\in_r$ = 6.91. If it operates in the dominant mode i.e. TE_{10} at 6 GHz.

Previous Years' Question

The standard air-filled rectangular waveguide of dimensions a = 2.29 cm for b = 1.02 cm is designed for radar applications.
It is desired that these waveguides operate only in the dominant TE_{10} mode but not higher than 95% of the next higher cut-off frequency. The range of the allowable operating frequency 'f' is:
a) 8.19 GHz ≤ f ≤ 13.1 GHz
b) 8.19 GHz ≤ f ≤ 12.45 GHz
c) 6.55 GHz ≤ f ≤ 13.1 GHz
d) 1.64 GHz ≤ f ≤ 0.24 GHz

Find the following:

i) Cut-off frequency.

ii) Phase velocity.

iii) Guided wavelength

Sol:

The given dimensions are 1.073 cm × 0.75 cm

For dominant mode i.e. TE_{10} mode, a ≥ 2b

$\therefore$ Let a = 1.073 cm

b = 0.75 cm

i) The cut-off frequency (f_c) will be:

$$(f_c)_{TE_{10}} = \frac{v_p}{2a} = \frac{c}{\sqrt{\epsilon_r}(2a)}$$

$$(f_c)_{TE_{10}} = \frac{3 \times 10^{10}}{\sqrt{6.91}(2 \times 1.073)} = \frac{3 \times 10^{10}}{(2.628)(2.146)}$$

$$= 0.532 \times 10^{10} = 5.320 \text{ GHz}$$

ii) The guided phase velocity is given as:

$$v_{pz} = \frac{v_p}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}} = \frac{c}{\sqrt{\epsilon_r}\sqrt{1-\left(\frac{f_c}{f}\right)^2}}$$

$$= \frac{3 \times 10^8/\sqrt{6.91}}{\sqrt{1-\left(\frac{5.320}{6}\right)^2}}$$

$$= \frac{1.141 \times 10^8}{\sqrt{1-0.786}} = \frac{1.141 \times 10^8}{\sqrt{0.214}}$$

$$v_{pz} = \frac{1.141 \times 10^8}{0.462} = 2.469 \times 10^8 \text{ m/s}$$

iii) The guided wavelength for guided medium:

$$\lambda_g = \frac{\lambda}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}}$$

Previous Years' Question

A 100 m section of an air-filled rectangular waveguide operating in the TE_{10} mode has a cross-sectional dimension of 1.071 cm × 0.5 cm. The pulse carrier of 21 GHz and 28 GHz are simultaneously launched at one end of the waveguide section. What is the time delay difference between the two pulses at the other end of the waveguide?

Previous Years' Question

If v is the velocity of propagation in an unbounded medium, v_p and v_g are the phase and groups velocities in a guide filled with a medium, having the same permittivity as that of the unbounded medium, then v, v_p and v_g are related as:

a) $v_p\ v = v_2$

b) $v_p\ v = v^2$

c) $v_p\ v_g = v^2$

d) $(v - v_p)(v - v_g) = v^2$

$$= \frac{v_p/f}{\sqrt{1-\left(\frac{5.320}{6}\right)^2}} = \frac{c}{f\sqrt{\epsilon_r}\sqrt{1-\left(\frac{f_c}{f}\right)^2}}$$

$$= \frac{3\times10^{10}}{\sqrt{6.91}\times6\times10^9\sqrt{1-\left(\frac{5.320}{6}\right)^2}}$$

$$\lambda_g = \frac{3\times10^{10}}{(2.628)\times(6\times10^9)(0.462)}$$

$$= \frac{3\times10^{10}}{7.284\times10^9} = 4.11\text{cm}$$

Example 3:

A 2.2 cm × 3.2 cm waveguide is filled with a dielectric material with ϵ_r = 4. If the waveguide operates at 20 GHz with TM_{11} mode, then determine:

i) Cut-off frequency.
ii) Phase constant.
iii) Phase velocity.

Sol:

The dimensions given are:

Let a = 3.2 cm, b = 2.2 cm

and operating frequency (f) = 10 GHz

i) The cut-off frequency is given by:

$$(f_c)_{TM_{11}} = \frac{c}{2\sqrt{\epsilon_r}}\sqrt{\left(\frac{m}{a}\right)^2+\left(\frac{n}{b}\right)^2}$$

$$= \frac{3\times10^{10}}{2\sqrt{4}}\sqrt{\left(\frac{1}{3.2}\right)^2+\left(\frac{1}{2.2}\right)^2}$$

$$= 0.75\times10^{10}\sqrt{(0.0976)+(0.2066)}$$

$$= 0.754\times10^{10}(0.5515) = 4.136\text{ GHz}$$

ii) The phase constant for guided medium β_g is given as:

Previous Years' Question

The dimension equation of a waveguide, which relates to wave-number to the frequency ω is:

$$K(\omega) = \frac{1}{c}\sqrt{\omega^2-\omega_0^2}$$

Where the speed of light c = 3×10^8 m/s and ω_0 is a constant.
If the group velocity is 2×10^8 m/s, then the phase velocity will be:

a) 1.5×10^8 m/s
b) 3×10^8 m/s
c) 4.5×10^8 m/s
d) 2×10^8 m/s

$$\beta_g = \beta\sqrt{1-\left(\frac{f_c}{f}\right)^2} = \frac{\omega}{v_p}\sqrt{1-\left(\frac{f_c}{f}\right)^2}$$

$$= \frac{2\pi f\sqrt{\epsilon_r}}{c}\sqrt{1-\left(\frac{4.136}{20}\right)^2}$$

$$= \frac{2\pi(20\times10^9)\sqrt{4}}{3\times10^8}\sqrt{1-\left(\frac{4.136}{20}\right)^2}$$

$$= 837.7580\sqrt{1-0.0427}$$

$$= 837.7580(0.9784)$$

$$\beta_g = 819.66 \text{ rad/m}$$

iii) The guided phase velocity is given by:

$$v_{pz} = \frac{v_p}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}}$$

$$= \frac{c}{\sqrt{\epsilon_r}\sqrt{1-\left(\frac{f_c}{f}\right)^2}} = \frac{3\times10^8}{\sqrt{4}\sqrt{1-\left(\frac{4.136}{20}\right)^2}}$$

$$= \frac{3\times10^8}{2(0.9784)} = 1.533\times10^8 \text{ m/s}$$

Intrinsic impedance for TE_{mn} mode:

- The intrinsic impedance is denoted by (η):

or $\eta_{TE} = \frac{E_x}{H_y} = \frac{-E_y}{H_x}\{\text{for } TE_{mn} \text{ mode}\}$

The ratio of electric field and magnetic field gives intrinsic impedance:

Here $E_{zs} = 0$, $H_{zs} \neq 0$

- As we know,

$$H_{zs} = H_0 \cos\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right)e^{-\gamma z}$$

And $E_{xs} = -\frac{\gamma}{h^2}\frac{\partial E_{zs}}{\partial x} - \frac{j\omega\mu}{h^2}\frac{\partial H_{zs}}{\partial y}$

Put $E_{zs} = 0$

Previous Years' Question

An air-filled rectangular waveguide has inner dimensions of 3cm × 2cm. The wave impedance of the TE_{20} mode of propagation in the waveguide at a frequency of 30 GHz is (free space impedance $\eta_0 = 377\Omega$) is:

a) 308 Ω **b)** 461 Ω
c) 355 Ω **d)** 400 Ω

Now, to get the value of H_{ys}, we know that-

$$H_{ys} = \frac{-\gamma}{h^2}\frac{\partial H_{zs}}{\partial y} - \frac{j\omega \in}{h^2}\frac{\partial E_{zs}}{\partial x}$$

$$H_{ys} = \frac{-\gamma}{h^2}\frac{\partial}{\partial y}\left[H_0 \cos\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right)e^{-\gamma z}\right]$$

$$H_{ys} = \frac{\gamma}{h^2}H_0\left(\frac{n\pi}{b}\right)\cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)e^{-\gamma z}$$

- The intrinsic impedance η_{TE} is given by:

$$\eta_{TE} = \frac{E_{xs}}{H_{ys}} = \frac{j\omega\mu}{h^2}$$

$$= \frac{\frac{j\omega\mu}{h^2}\left(\frac{n\pi}{b}\right)H_0\cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)e^{-\gamma z}}{\frac{\gamma}{h^2}H_0\left(\frac{n\pi}{b}\right)\cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)e^{-\gamma z}}$$

$$= \frac{j\omega\mu}{j\beta_g} = \frac{\omega\mu}{\beta_g} = \frac{\omega\mu}{\beta\sqrt{1-\left(\frac{f_c}{f}\right)^2}}$$

where $\frac{\omega\mu}{\beta} = \frac{\omega\mu}{\omega\sqrt{\mu\in}} = \sqrt{\frac{\mu}{\in}} = \eta$

Therefore, $\frac{\omega\mu}{\beta} = \eta$ = intrinsic Impedance for lossless medium.

Thus, $$\eta_{TE} = \frac{\eta}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}} \quad ...(7.47)$$

Intrinsic impedance for TM_{mn} mode:

- The intrinsic impedance is denoted by η_{TM} for this mode.
- In TM_{mn} mode, $E_{zs} \neq 0$ and $H_{zs} = 0$.

$$\therefore \ \eta_{TM} = \frac{E_{xs}}{H_{ys}}$$

Previous Years' Question

A rectangular hollow metal waveguide internal cross-sections of 7.366 cm × 3.55 cm carries a 3 GHz signal in the TE_{10} mode. Calculate the maximum power handling capability of the waveguide assuming the maximum permissible electric field inside the waveguide to be 30kV/cm.

As we know,

$$E_{zs} = E_0 \sin\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi y}{b}\right)e^{-\gamma z}$$

$$E_{xs} = \frac{-\gamma}{h^2}\frac{\partial E_{zs}}{\partial x} - \frac{j\omega\mu}{h^2}\frac{\partial H_{zs}}{\partial y}$$

here, $H_{zs} = 0$,

$$E_{xs} = \frac{-\gamma}{h^2}\frac{\partial}{\partial x}\left[E_{zs}\right]$$

$$= -\frac{\gamma}{h^2}E_0\left(\frac{m\pi}{a}\right)\cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)e^{-\gamma z}$$

$$H_{ys} = \frac{-\gamma}{h^2}\frac{\partial H_{zs}}{\partial y} - \frac{j\omega\in}{h^2}\frac{\partial E_{zs}}{\partial x}$$

$$H_{ys} = \frac{-j\omega\in}{h^2}\frac{\partial}{\partial x}(E_{zs})$$

$$= \frac{-j\omega\in}{h^2}E_0\left(\frac{m\pi}{a}\right)\cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)e^{-\gamma z}$$

Therefore,

$$\eta_{TM} = \frac{E_{xs}}{H_{ys}} = \frac{\gamma}{j\omega\in} = \frac{j\beta_g}{j\omega\in}$$

$$= \frac{\beta\sqrt{1-\left(\frac{f_c}{f}\right)^2}}{\omega\in}$$

$$\eta_{TM} = \frac{\omega\sqrt{\mu\in}}{\omega\in}\sqrt{1-\left(\frac{f_c}{f}\right)^2}$$

$$\eta_{TM} = \sqrt{\frac{\mu}{\in}}\sqrt{1-\left(\frac{f_c}{f}\right)^2} = \eta\sqrt{1-\left(\frac{f_c}{f}\right)^2}$$

Hence,

$$\eta_{TM} = \eta\sqrt{1-\left(\frac{f_c}{f}\right)^2} \qquad ...(7.48)$$

Previous Years' Question

The cut-off frequency of the TE_{01} mode of an air-filled rectangular waveguide having inner dimensions a cm × b cm (a > b) is twice that of the dominant TE_{10} mode. When the waveguide is operated at a frequency that is 25% higher than the cut-off frequency of the dominant mode, the guide wavelength is found to be 4 cm. The value of b (in cm, correct to two decimal places is _____.)

Rack Your Brain

Why circular waveguides are not used over rectangular waveguides?

where η = intrinsic impedance for lossless medium inside the waveguide.

Relationship between η_{TE} and η_{TM}:

$$\eta_{TE} \times \eta_{TM} = \frac{\eta}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}} \times \eta\sqrt{1-\left(\frac{f_c}{f}\right)^2}$$

$$\eta = \sqrt{\eta_{TE}\,\eta_{TM}} \qquad ...(7.49)$$

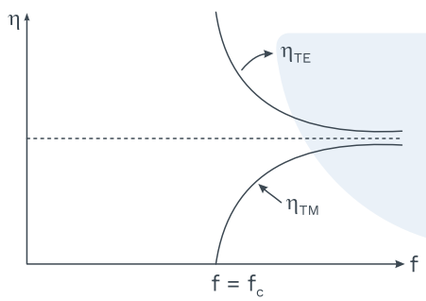


Fig. 7.6 Intrinsic Impedance vs Frequency

- For

$$\begin{cases} f = 0, & \eta_{TM} = 0 \\ f > f_c, & \eta_{TM} \text{ increases} \\ f < f_c, & \eta_{TM} \text{ decreases} \end{cases}$$

- For air filled waveguides:

$$\eta = \eta_0 = 120\pi$$

- For de-ionized or dielectric filled waveguide:

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = 120\pi\sqrt{\frac{\mu_r}{\epsilon_r}}$$

POWER FLOW IN THE WAVEGUIDE

- The poynting vector is used to evaluate the power flow in the waveguide:

The average poynting vector ($\vec{P}_{avg}$) is given by:

Rack Your Brain

Which mode in a circular waveguide has an attenuation effect decreasing with an increase in frequency?

Previous Years' Question

The phase velocity of electromagnetic wave propagation in a hollow metallic rectangular waveguide in the TE_{10} mode is:

a) Less than the velocity of light in a free space.
b) Equal to the velocity of the light in a free space.
c) Equal to its group velocity.
d) Greater than the velocity of the light in a free space.

$$(\vec{P}_{avg}) = \frac{1}{2}\text{Re}\left[\vec{E}_s \times \vec{H}_s^*\right]$$

Since the wave is travelling in the z-direction:

$E_s = (E_{xs}, E_{ys})$, $H_s = (H_{ys}, H_{xs})$

$$\vec{P}_{avg} = \frac{1}{2}\text{Re}\left[E_{xs}\hat{a}_x + E_{ys}\hat{a}_y\right]$$

$$\vec{P}_{avg} = \frac{1}{2}\text{Re}\left[E_{xs}\hat{a}_x + E_{ys}\hat{a}_y\right] \times \left[H_{xs}^*\hat{a}_x + H_{ys}^*\hat{a}_y\right]$$

$$= \frac{1}{2}\text{Re}\left[E_{xs}H_{ys}^*\hat{a}_z + E_{ys}H_{xs}^*(-\hat{a}_z)\right]$$

$$= \frac{1}{2}\text{Re}\left[E_{xs}H_{ys}^*\hat{a}_z - E_{ys}H_{xs}^*\right]\hat{a}_z$$

$$= \frac{1}{2}\text{Re}\left[E_{xs}\frac{E_{xs}^*}{\eta_{TE/TM}} + E_{ys}\frac{E_{ys}^*}{\eta_{TE/TM}}\right]\hat{a}_z$$

$$\vec{P}_{avg} = \frac{1}{2}\text{Re}\left[\frac{|E_{xs}|^2 + |E_{ys}|^2}{\eta_{TE/TM}}\right]\hat{a}_z \qquad \left\{\because \eta = \frac{E_{xs}}{H_{ys}} = \frac{-E_{ys}}{H_{xs}}\right\}$$

$\therefore$ Time average poynting vector will be:

$$\vec{P}_{avg} = \frac{1}{2}\left[\frac{|E_{xs}|^2 + |E_{ys}|^2}{\eta_{TE/TM}}\right]\hat{a}_z \qquad \ldots(7.50)$$

- For TE_{10} mode ($E_{zs} = 0$, $H_{zs} \neq 0$)

$$H_{zs} = H_0 \cos\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right)e^{-j\beta_g z}$$

$$H_{zs} = H_0 \cos\left(\frac{\pi x}{a}\right)e^{-j\beta_g z} \qquad \left[\because m = 1 \text{ and } n = 0\right]$$

$$E_{xs} = \frac{-\gamma}{h^2}\frac{\partial E_{zs}}{\partial x} - \frac{j\omega\mu}{h^2}\frac{\partial H_{zs}}{\partial y}$$

$$\therefore E_{xs} = \frac{-j\omega\mu}{h^2}\frac{\partial H_{zs}}{\partial y}$$

$$E_{xs} = \frac{-j\omega\mu}{h^2}\frac{\partial}{\partial y}\left[H_0 \cos\left(\frac{\pi x}{a}\right)e^{-j\beta_g z}\right]$$

$\therefore E_{xs} = 0$

Grey Matter Alert!

TM_{01}, TM_{02}, TM_{11}, TM_{12} are the possible TM modes in a circular waveguides.

Rack Your Brain

Why is TM_{01} mode preferred to the TE_{01} mode in a circular waveguide?

And $E_{ys} = \frac{-\gamma}{h^2}\frac{\partial E_{zs}}{\partial y} + \frac{j\omega\mu}{h^2}\frac{\partial H_{zs}}{\partial x}$

$$E_{ys} = \frac{j\omega\mu}{h^2}\frac{\partial H_{zs}}{\partial x}$$

$$E_{ys} = \frac{j\omega\mu}{h^2}\left[H_0\left(\frac{\pi}{a}\right)\sin\left(\frac{\pi x}{a}\right)e^{-j\beta_g z}\right]$$

- Substitute the values in equations 7.50):

$$\vec{P}_{avg} = \frac{1}{2\eta_{TE_{10}}}\left[0^2 + \left(\frac{\omega\mu}{h}H_0\left(\frac{\pi}{a}\right)\sin\left(\frac{\pi x}{a}\right)\right)^2\right]$$

$$\vec{P}_{avg} = \frac{\omega^2\mu^2 H_0}{2h^4\eta_{TE_{10}}}\left(\frac{\pi}{a}\right)^2\sin^2\left(\frac{\pi x}{a}\right)\hat{a}_z \qquad ...(7.51)$$

Since, m = 1 and n = 0

$$h^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 = K_x^2 + K_y^2$$

$$h^2 = \left(\frac{\pi}{a}\right)^2$$

$$\therefore \vec{P}_{avg} = \frac{\omega^2\mu^2 H_0^2}{2h^2\eta_{TE_{10}}}\sin^2\left(\frac{\pi x}{a}\right)\hat{a}_z$$

- Power, $P = \int_s \vec{P}_{avg}\cdot d\vec{S}$

$$= \int_{x=0}^{a}\int_{y=0}^{b}\vec{P}_{avg}\hat{a}_z\cdot dx\,dy\,\hat{a}_z$$

$$P = \int_{x=0}^{a}\int_{y=0}^{b}\frac{\omega^2\mu^2 H_0^2}{2h^2\eta_{TE_{10}}}\sin^2\left(\frac{\pi x}{a}\right)dx\,dy$$

- Here, $\int_{n=0}^{a}\sin^2\left(\frac{\pi}{a}x\right)dx$

$$= \int_{x=0}^{a}\frac{1-\cos\left(\frac{2\pi x}{a}\right)}{2}dx = \left[\frac{a}{2} - \frac{1}{2}\sin\left(\frac{2\pi x}{a}\right)\frac{a}{2\pi}\right]_0^a = \frac{a}{2}$$

$$\therefore P = \frac{\omega^2\mu^2 H_0^2}{2h^2\eta_{TE_{10}}}\left(\frac{a}{2}\right)(b)$$

Grey Matter Alert!

Circular waveguides are used as attenuators and phase-shifters

$$P = \frac{\omega^2\mu^2 H_0^2}{2h^2\eta_{TE_{10}}}\left(\frac{ab}{2}\right) = \frac{\omega^2\mu^2 H_0^2}{2\left(\frac{\pi}{a}\right)^2\eta_{TE_{10}}}\left(\frac{ab}{2}\right)$$

- Rearranging the above equation, we get:

$$P = \left(\frac{\omega\mu a}{\pi}H_0\right)^2\left(\frac{ab}{4\eta_{TE_{10}}}\right)$$

$$P = \frac{E_0^2}{4\eta_{TE_{10}}}ab \qquad ...(7.52)$$

- As we know, plane wave power is given by:

$$P = \frac{E_0^2}{2\eta} \times \text{Area}$$

where $E_0 = \frac{\omega\mu a H_0}{\pi}$ = Peak value of an electric field.

- For dominant mode, the power will be:

$$P = \frac{E_0^2}{4\eta_{TE_{10}}}ab$$

Example 4:

Consider an air-filled rectangular waveguide transporting energy in the dominant mode at the rate of 2mW. It has cross-sectional dimensions of a = 4cm and b = 2cm.

If the frequency of operation is 10GHz, then determine the peak value of an electric field in the given waveguide.

Sol:

Given dimensions are:

a = 4 cm

b = 2 cm

Power, P = 2 mW = 2×10^{-3}W

Frequency, f = 10 GHz

Power for dominant mode:

$$P = \frac{E_0^2}{4\eta_{TE_{10}}}ab$$

$$E_0 = \sqrt{\frac{4P\eta_{TE_{10}}}{ab}}$$

As, cut-off frequency is $f_c = \frac{c}{2a} = \frac{3 \times 10^{10}}{2 \times 4} = 3.75\,\text{GHz}$

$$\eta_{TE_{10}} = \frac{120\pi}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}}$$

$$= \frac{120\pi}{\sqrt{1-\left(\frac{3.75}{10}\right)^2}}$$

$$\eta_{TE_{10}} = 406.66\,\Omega$$

$$\therefore \quad E_0 = \sqrt{\frac{4 \times 2 \times 10^{-3} \times 406.66}{84 \times 2 \times 10^{-4}}}$$

$$E_0 = 63.77 \text{ V/m}$$

CIRCULAR WAVEGUIDE

- Difference between rectangular and circular waveguides are-
  - **i)** Rectangular waveguide is a hollow metallic tube of rectangular cross-section, whereas the circular waveguide is a hollow metallic tube of circular cross-section.
  - **ii)** Rectangular waveguides are generally used in microwave application because of their small size and circular waveguide are used in rotational coupling.
  - **iii)** The dominant mode of rectangular waveguide is TE_{10}, while circular waveguide is TE_{11}.

But the hollow metallic tube is made up of brass of copper in both waveguides.

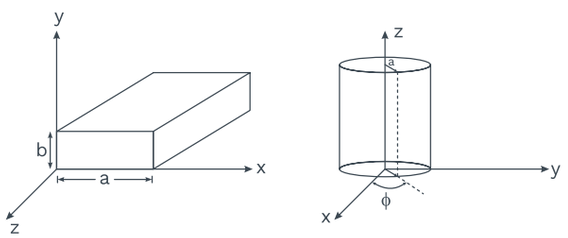


Fig. 7.7 Circular Waveguide Coordinates

Solution of the wave equation in cylindrical coordinate:

- The electric field and magnetic field wave equations are given by:

$$\nabla^2\vec{E} = \gamma^2\vec{E}$$

$$\nabla^2\vec{H} = \gamma^2\vec{H}$$

when $\gamma = \sqrt{j\omega\mu(\sigma + j\omega\in)} = \alpha + j\beta$

- The circular or cylindrical components of an electric field or magnetic fields satisfy the complex scalar wave equations or Helmholtz equation.
  $\nabla^2\psi = \gamma^2\psi$
- The scalar Helmholtz equation in cylindrical coordinates is given by:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(\frac{r\partial\psi}{\partial r}\right) + \frac{1}{r^2}\left(\frac{\partial^2\psi}{\partial\phi^2}\right) + \frac{\partial^2\psi}{\partial z^2} = \gamma^2\psi$$

- Using method of separation of variable (R, ϕ, Z)
  $\psi = R(r)\ \phi(\phi)\ Z(z)$

$$\frac{1}{r}\left(\frac{\partial}{\partial r}\frac{rR(r)\phi(\phi)Z(z)}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2R(r)\phi(\phi)Z(z)}{\partial\phi^2} + \frac{\partial^2R(r)\phi(\phi)Z(z)}{\partial z^2} = \gamma^2R(r)\phi(\phi)Z(z)$$

- In the above equation, divide by ψ on both sides.

$$\frac{1}{Rr}\frac{\partial}{\partial r}\left(\frac{r\partial R}{\partial r}\right) + \frac{1}{r^2\phi}\frac{\partial^2\phi}{\partial\phi^2} + \frac{1}{Z}\frac{\partial^2Z}{\partial z^2} = \gamma^2 \qquad \text{...(7.53)}$$

- Let $\frac{1}{Z}\frac{\partial^2Z}{\partial z^2} = \gamma_g^2$

$$\frac{\partial^2Z}{\partial z^2} = \gamma_g^2 Z$$

$$\frac{\partial^2Z}{\partial z^2} - \gamma_g^2 Z = 0$$

$(D^2 - \gamma_g{}^2)Z = 0$ (Auxiliary equation)

$D = \pm\gamma_g$

$$Z = Ae^{-\gamma_g z} + Be^{+\gamma_g z}$$

where γ_g = propagation constant in the waveguide.

- Now, substituting values in equation 7.53), we get:

$$\frac{1}{rR}\frac{\partial}{\partial r}\left(\frac{r\partial R}{\partial r}\right) + \frac{1}{r^2\phi}\frac{\partial^2\phi}{\partial\phi^2} + \left(\gamma_g^2 - \gamma^2\right) = 0$$

- Multiplication of r^2 on both sides of the above equation.

$$\frac{r}{R}\frac{\partial}{\partial r}\left(\frac{r\partial R}{dr}\right) + \frac{1}{\phi}\frac{\partial^2\phi}{\partial\phi^2} + \left(\gamma_g^2 - \gamma^2\right)r^2 = 0$$

Let $\frac{1}{\phi}\frac{\partial^2\phi}{\partial\phi^2} = -n^2$

$$\frac{\partial^2\phi}{\partial\phi^2} + n^2\phi = 0$$

$$(D^2 + n^2)\phi = 0$$

$$D = \pm nj$$

and $\phi = C \sin(n\phi) + D \cos(n\phi)$

- Now, substituting values in above equation:

$$\frac{r}{R}\frac{\partial}{r}\left(\frac{r\partial R}{\partial r}\right) - n^2 + (\gamma_g^2 - \gamma^2)r^2 = 0$$

- Now, multiplying by R on both sides, we get:

$$r\frac{\partial}{\partial r}\left(\frac{r\partial R}{\partial r}\right) - n^2R + \left(\gamma_g^2 - \gamma^2\right)r^2R = 0$$

$$r\frac{\partial}{\partial r}\left(\frac{r\partial R}{\partial r}\right) + (K_c^2r^2 - n^2)R = 0 \qquad ...(7.54)$$

- This is Bessel equation of order 'n' where $K_c^2 = \gamma_g^2 - \gamma^2$ is a characteristic equation of Bessel equation.
- Solution of Bessel equation 7.54) is given by:

$R = P\, J_n(K_cr) + Q\, N_n(K_cr)$ ...(7.55)

where $J_n(K_cr)$ = Bessel's function of the first kind, which represents standing wave of cos (K_cr) for r < a.

$N_n(K_cr)$ = Bessel's function of second kind which represents standing wave of sin(K_cr) for r > a.

a = radius of the cylinder

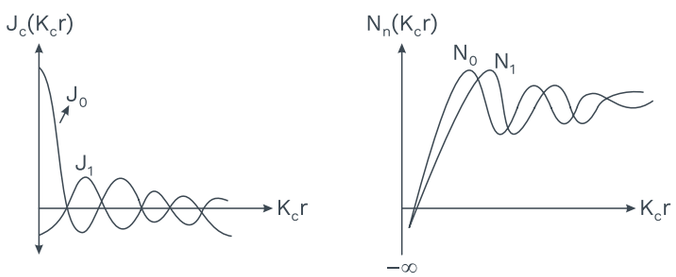


Fig. 7.8 Bessel's Function

Conclusion:

when r = 0, $N_n(0) = -\infty$

$J_n(0)$ = 1 (Maximum value)

The term $N_n(K_c r)$ is making whole equation 7.55) value as infinity ($-\infty$)
Thus, we will neglect the $N_n(K_c r)$.

- Total solution of Helmholtz equation:

$$\psi = \left[P\ J_n\left(K_c r\right) + Q N_n\left(K_c r\right)\right]\left[C\cos\left(n\phi\right) + D\sin\left(n\phi\right)\right]\left[e^{\gamma_g} + e^{-\gamma_g}\right]$$

At r = 0, $N_n \to \infty$, Thus,

$$\psi \to \infty$$

Thus, to makes it finite/proper, we have to take the constant Q → 0

$$\psi = P\,J_n(K_c r)\left[C\cos(n\phi) + D\sin(n\phi)\right]e^{-\gamma_g z}$$

Here, we are considering +z direction.

- Assuming lossless conditions:
  For lossless medium, $\alpha = 0$

$\therefore\ \gamma_g = +j\beta_g$

$\because$ By quadrature components,

$$C\cos\left(n\phi\right) + D\sin\left(n\phi\right) = \sqrt{C^2 + D^2}\cos\left(n\phi - \tan^{-1}\left(\frac{D}{C}\right)\right)$$

$$\psi = P\,J_n\left(K_c r\right)\sqrt{C^2 + D^2}\cos\left[n\phi - \tan^{-1}\left(\frac{D}{C}\right)\right]e^{-j\beta_g z}$$

$$\psi = \psi_0\,J_n(K_c r)\cos(n\phi)\,e^{-j\beta_g z} \qquad \ldots(7.56)$$

Here, neglecting phase of cos- terms.

i.e., $\tan^{-1}\left(\frac{D}{C}\right)$ (considering it very small)

$$\therefore\ \psi_0 = P\sqrt{C^2 + D^2}$$

- Also,

$$K_c^{\,2} = \gamma_g^{\,2} - \gamma^2$$

$$K_c^{\,2} = (j\beta_g)^2 - (j\beta)^2$$

$$K_c^{\,2} = -\beta_g^{\,2} + \beta^2$$

$$K_c^2 = -\beta_g^2 + \left(\omega\sqrt{\mu\in}\right)^2$$

$$K_c^2 = -\beta_g^2 + \omega^2\mu\in \qquad \ldots(7.57)$$

Rack Your Brain

Are there any elliptically shaped waveguides?

TE modes in circular waveguide:

- Considering the direction of wave propagation is z-direction in the TE mode,

$E_z = 0$ and $H_z \neq 0$

- The magnetic field in this mode will be:

$$H_{zs} = H_0 J_n(K_c r)\cos(n\phi)e^{-j\beta_g z}$$

- Applying the boundary conditions:

The tangential components of the electric field must be zero at the inner surface of the circular waveguide at r = a.

- As we know, the waveguide boundary is made up of metal and the electric field at metal (perfect conductor) is zero.
- To determine the values of constants, we now apply boundary conditions.
- At r = a (Boundary of circular waveguide)

$E_\phi = 0$ is the tangential component.

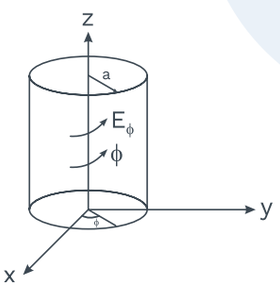


Fig. 7.9 Circular Waveguide

- The boundary and the electric field E_ϕ components are tangential to each other.
- As we know that-

$$E_\phi = \frac{j\omega\mu}{K_c^2}\frac{\partial H_{zs}}{\partial r}$$

$$E_r = \frac{-j\omega\mu}{K_c^2 r}\frac{\partial H_{zs}}{\partial \phi}$$

- If $E_\phi = 0$, then $\frac{\partial H_{zs}}{\partial r} = 0$

$$\therefore \quad \frac{\partial H_{zs}}{\partial r} = H_0 J_n'(K_c a)\cos(n\phi)e^{-j\beta_g z}$$

where $J_n'(K_c a) = \frac{\partial}{\partial r} J_n(K_c a)$

- Hence, to satisfy the above equation i.e.

Grey Matter Alert!

The dominant mode for a circular waveguide is defined as the lowest order mode having the lowest root value.

$$\frac{\partial H_{zs}}{\partial r} = 0$$

$\because$ we have to make $J'_n(K_c a) = 0$

- To change this Bessel function such that its value becomes zero.
  p^{th} order of $J'_n(K_c a)$ for TE_{np} modes.

Let $K_c a = X'_{np}$

p	n = 0	n = 1	n = 2	n = 3	n = 4
1	3.832	1.841	3.054	4.201	5.317
2	7.016	5.331	6.706	8.015	12.682
3	10.173	8.536	9.969	11.346	
4	13.324	11.706	13.170		

Table 7.3 p^{th} Root of nth Order Bessel Harmonics

where X'_{np} is a constant value

n = Number of input/feeds in ϕ direction

$(0 < \phi < \pi)$

p = Number of inputs feed points in radial directions.

Waveguide parameters

For TE_{np} mode:

As we know that:

$$K_c^2 = -\beta_g^2 + \omega^2 \mu \in$$

At $\omega = \omega_c$, $\beta_g = 0$

$$K_c^2 = \omega_c^2 \mu \in$$

$$\frac{X'_{np}}{a} = (2\pi f_c)\sqrt{\mu \in} \qquad \{\because X'_{np} = K_c a\}$$

$$f_c = \frac{X'_{np}}{2\pi a\sqrt{\mu \in}} \text{ or } \qquad f_c = \frac{X'_{np}\, v_p}{2\pi a} \qquad \ldots(7.58)$$

Equation 7.58) gives the formula for cut-off frequencies.

For air-filled waveguide:

$$v_p = c$$

$$f_c = \frac{X'_{np}\, c}{2\pi a}$$

$$f_c = \frac{X'_{np}(3\times 10^{10})}{2\pi a \text{ cm}}$$

{Here, 'a' is in cm and is a fixed quantity as it is a dimension}

- The minimum value of f_c denotes the dominant mode.
- From table 7.3), at n = 1 and p = 1, we are getting a minimum value of X'_{np}

∴ TE_{11} is a dominant mode.

- The cut-off frequency at (TE_{11}) for circular waveguide:

$$(f_c)_{TE_{11}} = \frac{1.841\times 3\times 10^{10}}{2\pi(a)} = \frac{0.879\times 10^{8}}{a}\text{Hz}$$

$$= \frac{8.79}{(a)}\text{GHz} \qquad \{a \text{ is in cm}\}$$

- Cut-off wavelength, $\lambda_c = \frac{v_p}{f_c} = \frac{2\pi a}{X'_{np}}$ ...(7.59)

For air-filled & dominant mode TE_{11}

$$\lambda_c = \frac{2\pi(a)}{1.841}$$

$\lambda_c = 3.41a$

- Phase velocity, $v_{pz} = \frac{\omega}{\beta_g}$

$$v_{pz} = \frac{v_p}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}} \qquad ...(7.60)$$

- The guided wavelength (λ_g):

$$\lambda_g = \frac{2\pi}{\beta_g} = \frac{\lambda}{\sqrt{1-\left(\frac{\lambda}{\lambda_c}\right)^2}} \qquad ...(7.61)$$

- The intrinsic impedance (η_{TE}):

$$\eta_{TE} = \frac{\eta}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}} \qquad ...(7.62)$$

Rack Your Brain

Which is the dominant mode out of TE_{11} or TM_{01} in a circular waveguide?

TM mode in circular waveguide:

For TM mode we know that:

$E_{zs} \neq 0$ and $H_{zs} = 0$

From equation 7.56),

$$E = E_0 J_n(K_c r)\cos(n\phi)e^{-j\beta_g z}$$

- At Boundary conditions-

when r = a, $E_{zs} = 0$

$$\therefore E_0 J_n(K_c a)\cos(n\phi)e^{-j\beta_g z} = 0$$

Hence, $J_n(K_c a) = 0$ will be zero if its argument will be zero.

Let argument of $J_n(K_c a) \Rightarrow X_{np} = K_c a$

p	n = 0	n = 1	n = 2	n = 3
1	2.405	3.832	5.136	6.380
2	5.520	7.106	8.417	9.761
3	8.645	10.173	11.620	13.015
4	11.792	13.324	14.796	

Table 7.4 p^{th} Root of nth Order Bessel Harmonics

- From table 7.4), we can conclude that if n = 0 and p = 1 will give the minimum value of cut-off frequency.

$\therefore$ The cut-off frequency for TM_{np} mode is:

$$(f_c)_{TM_{np}} = \frac{X_{np} v_p}{2\pi a} \qquad \ldots(7.63)$$

- The minimum mode in TM_{np} is TM_{01}.
- Intrinsic impedance for TM mode will be:

$$\eta_{TM} = \eta\sqrt{1-\left(\frac{f_c}{f}\right)^2} \qquad \ldots(7.64)$$

Degenerate mode:

- The modes having the same cut-off frequencies.
- When $X'_{np} = X_{np}$, then we will get the same cut-off frequency in circular waveguides.

From Table 7.3) and Table (7.4),

- We can conclude that some modes of TE_{np} have the same cut-off frequency.

or we can conclude that:

$TE_{0p} = TM_{1p}$

where p = 1, 2, 3, ...

e.g. $TE_{01} = TM_{10}$, $TE_{02} = TM_{20}$, $TE_{03} = TM_{30}$ } Degenerate modes

Evanescent mode:

- The mode that will not propagate in the waveguide or in other words, the mode will not be supported by the waveguide.
- From table 7.4), we can observe that p = 0 doesn't exist.

Or general way, the evanescent mode can be determined by:

TE_{n0} and TM_{n0}

where n = 0, 1, 2, ...

Example 5:

A TE_{11} mode is propagating through a CWG. The radius of the guide is 3 cm, and the guide contains an air dielectric.

i) Determine the cut-off frequency.

ii) The wavelength in the guide for an operating frequency of 3 GHz.

iii) Determine the wave impedance in the guide.

Sol:

Given that TE_{11} is the dominant mode, and dimension 'a' = 3 cm, $\epsilon_r = 1$ (air)

A) The cut-off frequency will be:

$$(f_c)_{TE_{11}} = \frac{1.84}{2\pi} \times \frac{3 \times 10^8}{3 \times 10^{-2}}$$

$$= \frac{1.84}{2\pi} \times 10^{10}$$

$$= 0.29 \times 10^{10}$$

$$(f_c)_{TE_{11}} = 2.92 \times 10^9 \text{ Hz}$$

$$(f_c)_{TE_{11}} = 2.92 \text{ GHz}$$

B) The guided wavelength (λ_g) is given as:

$$\lambda_g = \frac{\lambda}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} = \frac{\left(\frac{3 \times 10^8}{3 \times 10^9}\right)}{\sqrt{1 - \left(\frac{2.92 \times 10^9}{3 \times 10^9}\right)^2}}$$

$$\lambda_g = \frac{0.1}{\sqrt{1-0.947}}$$

$$\lambda_g = \frac{0.1}{\sqrt{0.0526}} = \frac{0.1}{0.229}$$

$\lambda g = 0.4359$

$\lambda g = 43.5$ cm

C) The wave impedance is given by:

$$\eta_{TE} = \frac{\eta}{1-\left(\frac{f_c}{f}\right)^2}$$

$$= \frac{120\pi}{\sqrt{1-\left(\frac{2.92\times10^9}{3\times10^9}\right)^2}}$$

$$= \frac{120\pi}{\sqrt{1-0.947}}$$

$$= \frac{120\pi}{\sqrt{0.0526}}$$

$$= \frac{120\pi}{0.229}$$

$\eta_{TE} = 1643.41\ \Omega$

Example 6:

Consider an air-filled circular waveguide having a radius of 2 cm and it has to carry energy at a frequency of 10GHz. Find all the TE_{np} or TM_{np} for which energy transmission is possible.

Sol:

In a circular waveguide, the condition for wave propagation is:

$f > f_c$

For TE_{np} mode, $f > \frac{X'_{np}\, v_p}{2\pi a}$

$\because$ For air-filled

$v_p = c$

$$f > \frac{X'_{np}\, c}{2\pi a}$$

$$f > \frac{X'_{np}\times 3\times 10^{10}}{2\pi\times 2}$$

If the $X'_{np} < 4.188$ then those modes will propagate in this given circular waveguide.

From table 7.3)
TE_{01}, $TE_{21,}$ and TE_{11} are the modes that satisfy the $X'_{np} < 4.188$ conditions.
Hence, they will propagate in circular waveguide (CWG).

- For TM_{np} mode:

Condition is $X_{np} < 4.188$

From table 7.4)
TM_{01} and TM_{11} modes satisfy the necessary condition.
Hence, overall TE_{01}, TE_{21}, TE_{11}, $TM_{01,}$ and TM_{11} will propagate, and the energy transmission is possible in these modes only.

Example 7:
An air-filled circular waveguide is designed to operate at a frequency of 6GHz and has dimensions such that $f_c = 0.8f$, for the dominant mode. Determine:
i) The diameter of the guide.
ii) The waveguide wavelength & phase velocity in the guide.

Sol:
i) The given cut-off frequency:

$$(f_c)_{TE_{11}} = \frac{1.841 \times c}{2\pi a}$$

$$\text{Diameter, } d = \frac{1.841 \times 3 \times 10^{10}}{\pi \times 0.8 \times 6 \times 10^{3}} = 3.66\,\text{cm}$$

ii) The guided wavelength is given by:

$$\lambda_g = \frac{\lambda}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}} = \frac{\frac{3 \times 10^8}{6 \times 10^9}}{\sqrt{1-(0.8)^2}}$$

$\lambda_g = 8.33$ cm
The phase velocity in the waveguide (v_{pz}):

$$v_{pz} = \frac{v_p}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}} = \frac{3 \times 10^8}{\sqrt{1-(0.8)^2}}$$

$v_p = 5 \times 10^8$ m/s

Example 8:
An air-filled circular waveguide of 3 cm inside radius is operated in the TE_{01} mode. Evaluate:
i) The cut-off frequency.
ii) If the guide is to be filled with a dielectric material of $\epsilon_r = 2.5$, then up to what value its radius must be changed in order to maintain the cut-off frequency at its original value?

Sol:
i) The cut-off frequency (f_c) for TE_{01} mode:

$$(f_c)_{TE_{01}} = \frac{X'_{np} \times c}{2\pi a}$$

$$= \frac{X'_{01} \times 3 \times 10^{10}}{2\pi \times 3}$$

$$= \frac{3.832 \times 3 \times 10^{10}}{6\pi}$$

$$(f_c)_{TE_{01}} = 6.098 \times 10^9 \text{ Hz}$$

$$(f_c)_{TE_{01}} = 6.09 \text{ GHz}$$

ii) The radius must be:

$$(f_c)_{TE_{01}} = \frac{X'_{np} \times V_p}{2\pi a}$$

$$6.09 \times 10^9 = \frac{3.832 \times c}{2\pi \times a}$$

$$a = \frac{3.832 \times 3 \times 10^8}{6.09 \times 10^9 \sqrt{2.5}} \times \frac{1}{2\pi}$$

$$a = 1.9 \text{ cm}$$

Example 9:
An EM wave is propagating in TE_{11} mode through a circular waveguide. The diameter of the guide is 10 cm and the guide is air-filled. Find:
i) Cut-off frequency.
ii) Guided wavelength λ_g at 3 GHz.
iii) Wave impedance in the guide.

Sol:

i) The cut-off frequency at TE_{11} mode is given by:

$$f_c = \left(\frac{8.79}{a}\right) GHz = \frac{8.79}{10} GHz = 0.879\, GHz.$$

ii) The guided wavelength at 3GHz.

$$\lambda_g = \frac{\lambda}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}} = \frac{\left(\frac{3\times 10^{10}}{3\times 10^9}\right)}{\sqrt{1-\left(\frac{0.879}{3}\right)^2}}$$

$\lambda_g = 10.459$ cm

iii) The wave impedance is given by:

$$\eta_{TE_{11}} = \frac{\eta}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}} = \frac{377}{\sqrt{1-\left(\frac{0.879}{3}\right)^2}}$$

$= 394.305\ \Omega$

Chapter Summary



- A waveguide is a hollow metallic tube structure that is used to carry microwave energy from one place to another.
- In TEM mode, the electric field, the magnetic field and the direction of propagation are perpendicular to each other.
- In TE mode, the electric field will exist in the transverse (perpendicular) directions to the direction of propagation of wave only.
- In TM mode, the magnetic field will exist in the transverse direction to the direction of propagation.
- The transverse propagation constant is given by:

  $h^2 = \gamma^2 + \omega^2\mu\epsilon = K_x^2 + K_y^2$

  $\gamma^2 = \sqrt{K_x^2 + K_y^2 - \omega^2\mu\epsilon}$ and we know that $\gamma = \alpha + j\beta$

- **Cut-off frequency:** The minimum frequency above which wave propagates and below which the wave attenuates:

  $$f_c = \frac{c}{2}\sqrt{\left(\frac{m}{2}\right)^2 + \left(\frac{n}{b}\right)^2}$$

- **Cut-off wavelength:** Maximum wavelength above which wave will attenuate and below which the wave propagates:

  $$\lambda_c = \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}}$$

- **Dominant mode:** The mode that gives the minimum cut-off frequency and maximum cut-off wavelength. In a rectangular waveguide, TE_{01} is the dominant mode and in a circular waveguide TE_{11} is the dominant mode.
- **Guided phase constant (β_g):** If a wave travels in a waveguide (i.e., in guided medium) will be referred to as guided phase constant:

  $$\beta_g = \beta\sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

- Guided wavelength (λ_g) is given by $\frac{1}{\lambda^2} = \frac{1}{\lambda_c^2} + \frac{1}{\lambda_g^2}$

- Guided phase velocity (v_{pz}) is given by: $v_{pz} = \dfrac{v_p}{\sqrt{1-\left(\dfrac{f_c}{f}\right)^2}}$
- Guided group velocity (v_g) is given by: $v_g = \dfrac{\partial\omega}{\partial\beta_z} = v_p\sqrt{1-\left(\dfrac{f_c}{f}\right)^2}$
- Relation between v_p and v_g is given by:

  $v_p = \sqrt{v_{pz}\, v_g}$

  where v_p = phase velocity of plane EM wave in free space
  v_{pz} = Guided phase velocity
  v_g = Guided group velocity
- Intrinsic Impedance for TE_{mn} mode: $\eta_{TE} = \dfrac{\eta_0}{\sqrt{1-\left(\dfrac{f_c}{f}\right)^2}}$ $(\because\ \eta_0 = 120\pi = 377\Omega)$
- Intrinsic Impedance for TM_{mn} mode: $\eta_{TM} = \eta_0\sqrt{1-\left(\dfrac{f_c}{f}\right)^2}$
- Relationship between η_{TE} and η_{TM} is given as- $\eta = \sqrt{\eta_{TE}\,\eta_{TM}}$
- The circular waveguide (CGW) is a hollow metallic tube of circular cross-section
- Cut-off frequency

  $\left(f_c\right)_{TE_{11}} = \dfrac{8.79}{a}$ GHz {where 'a' is in cm}
- In the circular waveguide, the cut-off wavelength, $\lambda_c = \dfrac{v_p}{f_c} = \dfrac{2\pi a}{X'_{np}}$
- In the circular waveguide, for air-filled & dominant mode TE_{11}:

  $\lambda_c = \dfrac{2\pi(a)}{1.841}$ or $\lambda_c = 3.41a$
- In the circular waveguide, the phase velocity, $v_{pz} = \dfrac{\omega}{\beta_g}$ or $v_{pz} = \dfrac{v_p}{\sqrt{1-\left(\dfrac{f_c}{f}\right)^2}}$

- The guided wavelength $\lambda_g = \frac{2\pi}{\beta_g} = \frac{\lambda}{\sqrt{1-\left(\frac{\lambda}{\lambda_c}\right)^2}}$
- The intrinsic impedance $\eta_{TE} = \frac{\eta_0}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}}$
- Intrinsic impedance for TM mode will be, $\eta_{TM} = \eta_0\sqrt{1-\left(\frac{f_c}{f}\right)^2}$
- **Degenerate mode:** The modes having the same cut-off frequencies.

  e.g. $TE_{01} = TM_{10}$, $TE_{02} = TM_{20}$, $TE_{03} = TM_{30}$ } Degenerate modes for circular waveguide
- **Evanescent mode:** The mode that will not propagate in the waveguide or in other words, the mode will not be supported by the waveguide. And it can be determined by-
  TE_{n0} and TM_{n0} where n = 0, 1, 2, ...

PRACTICE QUESTIONS

Q1 A 6.7 cm × 4.9 cm air-filled waveguide is considered. What modes are possible if the operating frequency of 6 GHz is to be propagated down the waveguide?

a) TE_{01} b) TE_{20} c) TE_{10} d) TE_{02}

Q2 If a tunnel is modelled as an air-filled rectangular metallic waveguide with dimensions 'a' = 6.4m and 'b' = 12.8m. Then, the tunnel will pass:

a) a 8.75 MHz broadcast signal
b) a 125 MHz FM broadcast signal
c) a 25 MHz broadcast signal
d) None of the above

Q3 The rectangular waveguide is operating at 5GHz cut-off frequency with TE_{10} mode and at 12 GHz with TE_{01} mode.
What will be the cut-off frequency for the TE_{11} mode if the waveguide is filled with a lossless material having $\in_r = 2.25$ and $\mu_r = 1$ is:

a) 7.23 GHz b) 8.66 GHz c) 16.23 GHz d) 17.32 GHz

Q4 A rectangular waveguide (air-filled) is 150m in length and is terminated by the metal plate at the one end.
Assume that if a short part of a frequency 7.2 GHz is introduced at the input end. Then the time taken by the pulse to return back to the input end will be:
(Let the cut-off frequency is 6.5 GHz)

a) 1.29 μsec b) 2.325 μsec c) 4.29 μsec d) 1.13 μsec

Q5 **Calculate the dimensions of an air-filled rectangular waveguide for which the cut-off frequencies for TM_{11} and TE_{03} modes are both equal to 12 GHz. At 8GHz, determine the type of mode that will propagate in the waveguide from the following:**

a) Dominant mode will propagate.
b) Evanescent mode in the waveguide.
c) Degenerate mode
d) None of the above.

Q6 **An air-filled rectangular waveguide operates at 40 GHz. If the cut-off frequency of the TE_{12} mode is 25 GHz. Calculate the phase constant, phase velocity, and intrinsic impedance of this mode:**

a) 654.49 rad/m, 3.84×10^8 m/s, 483.9 Ω
b) 3.841 rad/m, 3.84×10^8 m/s, 400 Ω
c) 483.49 rad/m, 3.84×10^8 m/s, 493.8 Ω
d) 600 rad/m, 3.84×10^8 m/s, 243 Ω

Q7 **If the rectangular waveguide (air-filled) is required to be constructed to operate at 4 GHz in the dominant mode. Determine the waveguide dimensions 'a' and 'b' such that the operating frequency must be 12% higher than the cut-off frequency for the dominant mode (TE_{10}) and 15% below the cut-off frequency of the next higher mode?**

a) a = 4.2 cm; b = 3.187 cm
b) a = 3.4 cm; b = 3.187 cm
c) a = 8.4 cm; b = 6.375 cm
d) a = 4.2 cm; b = 6.375 cm

Q8 **For an air-filled rectangular waveguide, a TE_{mode} operating at 6 GHz has**

$$E_y = 5\sin\left(\frac{2\pi x}{a}\right)\cos\left(\frac{\pi y}{b}\right)\sin(\omega t - 12z)\,V/m$$

The mode of operation and intrinsic impedance are respectively:

a) TE_{12}, 3978.4 Ω
b) TE_{21}, 3816 Ω
c) TE_{22}, 2778.16 Ω
d) TE_{21}, 3978.4 Ω

Q9 **The rectangular waveguide with dimensions 'a' and 'b' as 2.286 cm and 1.016 cm, respectively. Also the y-component of the electric field for the TE mode is given by:**

$$E_y = \sin\left(\frac{3\pi x}{a}\right)\cos\left(\frac{3\pi y}{b}\right)\sin\left(10\pi \times 10^{10} t - \beta z\right) Vm^{-1}$$

The propagation constant is:

a) j400.7 m^{-1} **b) j800.14 m^{-1}**

c) j407.7 m^{-1} **d) j419.17 m^{-1}**

Q10 **A rectangular waveguide (1 cm × 3 cm) is filled with air is operating in the TE_{12} mode at a frequency higher than 20% of the cut-off frequency:**
The phase and group velocities are respectively:

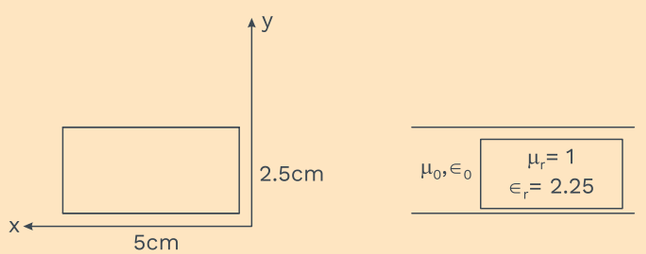
a) 5 × 10^8 m/s, 2 × 10^8 m/s **b) 5.42 × 10^8 m/s, 1.66 × 10^8 m/s**

c) 6.24 × 10^8 m/s, 1.66 × 10^8 m/s **d) 5.13 × 10^8 m/s, 1.42 × 10^8 m/s**

Q11 **A microwave transmitter antenna is connected by an air-filled waveguide of 2.5 cm × 1 cm dimensions. The ratio of group velocity to the medium velocity for transmission at 11 GHz is:**

a) 0.838 **b) 1.2** **c) 0.712** **d) 0.938**

Q12 **The rectangular waveguide with a cross-section as shown in the given figure has a discontinuous dielectric medium. The standing wave ratio if the guide operates at 8 GHz in the dominant mode:**



a) 2.566 **b) 1.56** **c) 1.93** **d) 4.13**

Q13 The cut-off frequency of a certain TE_{mn} mode is 16 GHz in a rectangular waveguide filled with polyethylene ($\in_r = 2.25\in_0$) and is operating at 24 GHz. The group velocity and intrinsic impedance of the mode are:

a) 1.49×10^8 m/s; 337.2 Ω
b) 1.23×10^6 m/s; 2.31 Ω
c) 1.49×10^8 c/s; 1.377 Ω
d) 1.31×10^6 m/s; 337 Ω

Q14 The ratio of area of circular waveguide to standard rectangular waveguide having same dominant mode cut-off frequency is:

a) 0.3183 b) 2.1581 c) 0.2196 d) 3.14

Q15 For TM_{01} mode of propagation in a circular waveguide with $X_{01} = 2.405$, with the inner diameter of the circular waveguide being equal to 20 mm. Find the cut off frequency for this mode of propagation?

a) 4.593 GHz b) 5.741 GHz c) 5.021 GHz d) 3.167 GHz

Answer key:

Q1	a), b) and c)	Q2	b) and c)	Q3	b)
Q4	a)	Q5	a)	Q6	a)
Q7	a)	Q8	d)	Q9	a)
Q10	a)	Q11	a)	Q12	b)
Q13	a)	Q14	b)	Q15	b)